

Built Environment Impacts on Zonal E-Scooter Expenses: A Bayesian Machine Learning Approach

Abstract

Shared e-scooters are reshaping urban mobility, yet zonal total trip expense patterns, a key to operator viability and transport equity, remain unexplored. This study examines how built environment factors affect zonal e-scooter trip expenses based on Chicago data, using a novel lognormal regression model enhanced by Bayesian Additive Regression Trees (LN + BART) to accommodate the right-skewed distribution of the trip expenses. The model outperforms traditional approaches by better capturing the nonlinear relationship. Partial dependence plots are used to visualize the results. Areas with moderate incomes, high POI density, and CBD proximity yield disproportionately higher revenues. However, zones with many car-free households are associated with lower revenues, highlighting affordability barriers despite clear mobility needs. This research advances transport economics by combining distribution-aware modeling with Bayesian machine learning, enhancing prediction and interpretability. It also offers practical tools for operators to optimize deployment and for policymakers to design targeted equity subsidies.

Keywords: micro-mobility; Bayesian Additive Regression Trees; nonlinear effects; spatial analysis

1. Introduction

Shared electric scooters (e-scooters) emerged as a transformative urban mobility innovation in September 2017 in Santa Monica, California, and rapidly expanded across the United States with 86 million trips by 2019 (NACTO, 2019). This unprecedented adoption rate stems from e-scooters' capacity to provide environmentally sustainable transportation (Weschke et al., 2022) while addressing critical first-and-last-mile connectivity challenges (Zuniga-Garcia et al., 2022).

While numerous studies have investigated usage patterns (Christoforou et al., 2021; Dibaj et al., 2021), consumer perception (Kopplin et al., 2021; Yang et al., 2021), and environmental impacts (De Bortoli and Christoforou, 2020; Moreau et al., 2020), zonal trip expenses (from the users' perspective) or revenues (from the operator's perspective), a fundamental economic indicator directly impacting both user adoption and service provider sustainability, remain insufficiently examined. Unlike ridership metrics that focus solely on the number of trips, the expense of trip is calculated considering both the trip distance and the duration. Thus, the total expense generated from a zone takes into consideration the number of trips, trip distance, and trip duration, which makes it a more comprehensive indicator of the traveler's demand. Besides, the spatial distribution of this indicator also reveals where revenue is generated, which is helpful for the operator to designate the operation area.

To study the characteristics of areas with high zonal e-scooter ridership revenues, the relationship between the dependent variable, zonal e-scooter sharing trip expenses, and the independent variables, built environment attributes of the zones, need to be constructed. However, the zone-level expense usually has a non-negative and skewed distribution, which violates the assumption of traditional modeling approaches that the response variable should follow the normal distribution, leading to potentially biased parameter estimations. Thus, specialized statistical frameworks that can accurately model the underlying data generation process are needed. Without proper adaptation to these distributional characteristics, even sophisticated machine learning methods may fail to capture the true relationship between built environment factors and zone-level expense.

To sum up, while the zone-level expense could both reflect the e-scooter sharing users' and the operators' profitability, there are very limited studies focusing on this indicator. Thus, this study aims to investigate the relationship between built environment and the zone-level expense to help operators determine the areas with both high user demand and high revenue, and to reveal the characteristics of these areas so that the

operator could identify such areas when implementing the e-scooter sharing service in other cities. The 2019 e-scooter trip data in Chicago is used in this study to demonstrate this point.

The remainder of the paper is structured as follows: Section 2 conducts a comprehensive literature review on the relationship terms of the contributing factors and modeling approaches. Section 3 describes the data preparation process, followed by the methodology section. Section 5 presents and discusses the modeling results. Finally, this paper concludes in Section 6 with a summary of key findings, methodological contributions, practical implications, and directions for future research.

2. Literature review

2.1. Built environment factors influencing e-scooter sharing ridership

There have been many studies performed on the relationship between built environment variables and e-scooter sharing ridership. These studies identify three primary categories of factors: transportation infrastructure, land use patterns, and socio-demographic characteristics.

First, transportation infrastructure elements create the foundational network upon which e-scooter services operate, directly influencing usage intensity and trip characteristics. Research consistently identifies intersection density, roadway connectivity, and transit infrastructure as critical determinants of e-scooter usage (Huo et al., 2021; Yang et al., 2022). The relationship between e-scooters and public transit reveals important complementary effects rather than competitive dynamics. Contrary to intuitive expectations, even bikeshare infrastructure stimulates rather than suppresses e-scooter travel (Yang et al., 2021), suggesting that micro-mobility options collectively expand the active travel market. Protected bicycle infrastructure that enhances rider safety significantly increases usage rates (Jin et al., 2023), demonstrating how infrastructure quality, not merely presence, influences utilization patterns and subsequent revenue generation.

Regarding land use patterns, downtown districts, university campuses, and mixed-use developments consistently emerge as high-demand zones for e-scooter services due to their trip-generating characteristics (Caspi et al., 2020; Huo et al., 2021; Tuli et al., 2021). The diversity and intensity of land use significantly influence e-scooter usage. Areas combining commercial, educational, and residential functions are associated with higher ridership (Bai and Jiao, 2020). Proximity to recreational facilities and open spaces are positively correlated with demand (Caspi et al., 2020; Tuli et al., 2021), while distance from the central business district is negatively correlated with ridership (Jin et al., 2023). These findings highlight the important influence of centrality and land use diversity on micro-mobility demand distribution.

Among the socio-demographic characteristics, population density, employment patterns, vehicle ownership, racial composition, and income distribution consistently emerge as significant predictors of e-scooter usage (Badia and Jenelius, 2023; Caspi et al., 2020; Tuli et al., 2021). Areas with higher proportions of younger residents and students demonstrate increased ridership, reflecting the technology adoption patterns of these populations (Bai and Jiao, 2020; Hosseinzadeh et al., 2021), while income distribution exhibits complex relationships with e-scooter usage: moderate-to-high income neighborhoods often show stronger adoption rates despite the relatively affordable nature of these services (Tuli et al., 2021). Neighborhoods with higher percentages of car-free households have higher e-scooter ridership, highlighting these services' role in expanding mobility options where private vehicle ownership is limited (Jin et al., 2023; Tuli et al., 2021). Communities with higher public transit reliance also show increased e-scooter usage (Badia and Jenelius, 2023; Caspi et al., 2020; Tuli et al., 2021), further confirming the complementary relationship between micro-mobility and conventional transit services (Tuli et al., 2021).

2.2. Modeling approach

Zone-level trip expense data, like ridership data, exhibits distinctive statistical properties that influence the choice of modeling approach. Both variables are non-negative, typically right-skewed, and exhibit heteroscedastic. Their similar distributional characteristics make the modeling approach designed for analyzing ridership also suitable for studying zone-level trip expense. Previous studies employed many nonlinear methods to investigate the relationship between built environment factors and e-scooter ridership. These include the multinomial logit model (Lazarus et al., 2020), geographically weighted regression model (Caspi et al., 2020), random-effects negative binomial model (Tuli et al., 2021), generalized additive model (Hosseinzadeh et al., 2021) and multilevel mixed-effects negative binomial model (Huo et al., 2021). Recent studies used machine learning models to investigate the nonlinear relationship. For example, gradient boosting decision tree (GBDT) and random forest (RF) are widely used. (Yang et al., 2022) employed GBDT to explore the nonlinear effects of the built environment on e-scooter sharing ridership in Los Angeles. Similarly, (Jin et al., 2023) utilized GBDT to investigate the relationship between built environment factors and e-scooter sharing volume on the road segments. (Silveira-Santos et al., 2024) used RF to forecast travel demand for e-scooter service in Madrid. While traditional machine learning approaches (e.g., GBDT and RF) can effectively capture nonlinear relationships, they present a critical limitation in analyzing the right-skewed expense data in this study, where many conventional methods require substantial preprocessing for non-normal distributions and often yield unrealistic predictions for inherently non-negative variables like the trip expenses.

The Bayesian Additive Regression Trees (BART) model (Chipman et al., 2010b) is a novel Bayesian machine learning method to understand the complex nonlinear relationships in different domains, such as transportation (Li et al., 2022), environment (Janizadeh et al., 2021; Zhang et al., 2020), and economy (Prüser, 2019). However, these studies set the prior for BART as the default normal regression, which overlooks one key distinction between the Bayesian models and ensemble models: Bayesian models can flexibly set priors based on the distribution of the variables. Due to its flexibility in handling different distributions of outcome variables (Martin, 2024), Bart has been extended to survival regression (Sparapani

et al., 2016) and log-linear regression (Murray, 2021) to better accommodate non-normal outcomes, where the link function of outcomes is assumed to follow normal distributions. For example, (Murray, 2021) detailed the log-linear BART for multinomial logistic and count regression models, characterized by better prediction performance and understanding of data generation processes.

3. Data preparation

3.1. Study area

This study focuses on Chicago, the third largest city in the United States, with a population exceeding 2.7 million and a gross domestic product (GDP) of \$725.75 billion in 2023 (Statista, 2024). Chicago represents an ideal study context due to its e-scooter pilot program launched by the Chicago Department of Transportation in 2019, which was specifically designed to foster a more sustainable and equitable transportation system (NACTO, 2019). The analysis concentrates on the pilot program's designated deployment area covering the northwest and west sides of the city, encompassing 249 of Chicago's 801 census tracts while excluding the CBD, as illustrated in Figure 1. Census tracts were selected as the spatial unit of analysis because they provide more granular location information than larger community areas, enabling more detailed spatial examination while maintaining broader applicability to other cities utilizing census tract-based data.



Figure 1. Study area in Chicago

3.2. Data collection and processing

The analytical framework for this study integrates multiple data sources via systematic collection, validation, and transformation. From the Chicago Data Portal (CDP, 2023), we extracted 710,839 e-scooter trip records covering June 15 to October 15, 2019. Each record contained comprehensive trip attributes: unique identifiers, temporal information (start/end times), spatial coordinates (origin/destination locations), and trip metrics (distance traveled and duration). To ensure data integrity, we implemented a rigorous filtering process that eliminated trips with speeds exceeding 31.1 miles per hour (50 km/h) (Caspi et al., 2020), distances shorter than 87.5 yards (80 m), and durations exceeding 7,200 seconds (McKenzie, 2019).

This validation procedure yielded 562,947 usable trips, representing 79.2% of the original dataset. For each valid trip, we calculated its cost using Chicago's standard pricing model: a \$1 base fee plus \$0.15 per minute of usage (Freund, 2019). Trip origins were selected as the primary spatial reference point as they more accurately reflect the initial demand for e-scooters within specific areas. The dependent variable—total trip expense per acre—was calculated by aggregating all trip costs originating within each census tract and normalizing by tract area to enable cross-spatial comparison.

The built environment can be categorized into three groups: transportation infrastructure, land use, and socio-demographics, which serve as the explanatory variables of this study. Transportation infrastructures, including bus routes, road networks, and Chicago's bike-sharing system (Divvy, 2023), were obtained from the CDP website and imported into QGIS for density calculations using spatial intersections. Land use data were assembled from multiple sources: Points of Interest (POI) from OpenStreetMap (EPA, 2023), land use classifications from the Chicago Metropolitan Agency for Planning (CMAP, 2023), and distance to CBD calculated using QGIS's centroids and distance matrix functions. The mixed land-use entropy was calculated using Equation 1 (Jin et al., 2023):

$$Entropy = \begin{cases} 0, & k = 1 \\ -\frac{[\sum_{j=1}^k P_j \ln(P_j)]}{\ln(k)}, & k \geq 2 \end{cases} \#(1)$$

where P_j represents the proportion of land use type j (where j ranges from 1 to 6) within the census tract, and k indicates the total number of land use types in the study area. The entropy value ranges from 0 to 1, where 0 indicates the presence of only one type of land use, and values closer to 1 indicate higher diversity in land use types, with 1 representing a perfectly balanced distribution of all land use types (Song et al., 2013). Socio-demographic variables were derived from the 2020 American Community Survey (ACS) 5-Year Estimates and the 2018 Census ACS, providing data on population characteristics, vehicle ownership patterns, employment status, racial composition, and income metrics. For safety considerations, crime

density was calculated using incident data from the CDP. This multi-source approach required careful harmonization of spatial references, temporal frames, and measurement scales to create a comprehensive analytical database at the census tract level.

3.3. Descriptive analysis

The response variable (total trip expense per acre) exhibited pronounced right-skew as illustrated in Figure 2, with values ranging from 0 to 634.8 and a standard deviation (5461.8) substantially exceeding the mean (29.73). This distributional characteristic violates normality assumptions underlying traditional regression approaches, directly informing our methodological decision to employ lognormal regression enhanced by Bayesian Additive Regression Trees. Table 1 presents descriptive statistics for all variables, including means, standard deviations, and range values. Several transportation infrastructure variables show considerable variation across the study area, with Divvy trips density ranging from 0 to 94.65 trips per acre (mean = 6.41) and road density spanning from 11.61 to 52.10 miles per acre (mean = 31.24). Land use metrics demonstrate similar heterogeneity, with distance to CBD varying from 1.13 to 18.1 miles (mean = 8.08) and POI density ranging from 0 to 0.92 points per acre (mean = 0.12). Socio-demographic variables reflect Chicago's diverse urban fabric, with car-free households ranging from 2% to 69% across census tracts (mean = 25%) and the proportion of white residents varying from 0% to 89% (mean = 28%).

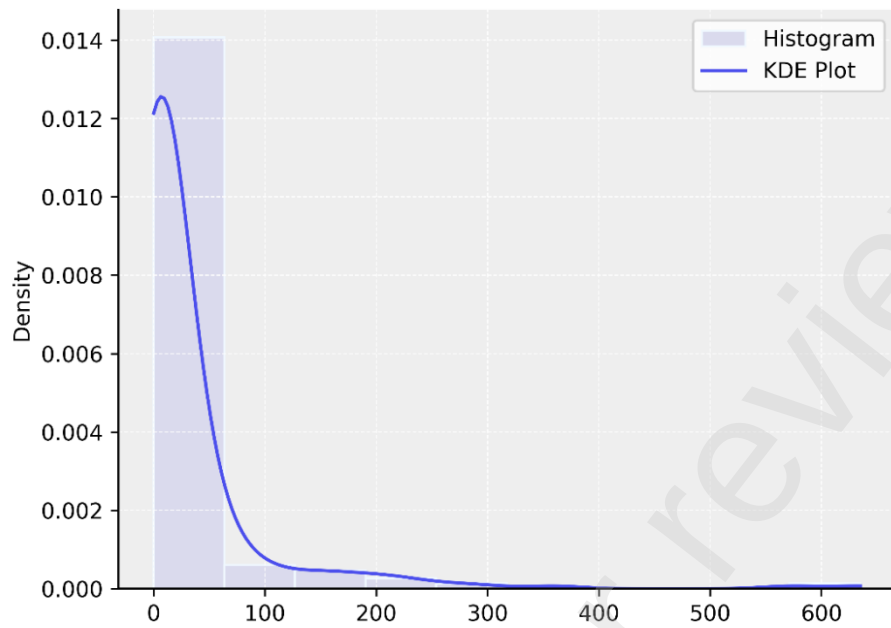


Figure 2 The distribution of total trip expense and its KDE curve

Table 1. Descriptive statistics of variables

Variables	Description	Source	Mean	S.D.	Min	Max
Response Variables						
Total trip expense	Total trip expense per acre	CDP	29.73	5461.8	0	634.8
Explanatory Variables						
<i>Transportation infrastructure</i>						
Divvy docks density	Density of Divvy docks	CDP	0.15	0.02	0	0.72
Divvy trips density	The density of Divvy trips	CDP	6.41	174.84	0	94.65
Bus stop density	Density of bus stops	CDP	0.10	0.01	0.02	0.26
Road density	Density of total road length	CDP	31.24	43.04	11.61	52.10

<i>Land use</i>						
Distance to CBD	Distance from the center of CT to the center of CBD	CDP	8.08	10.41	1.13	18.1
POI density	Density of POI	OSM	0.12	0.02	0	0.92
Commercial	Density of commercial area	CMAP	0.13	0.005	0	0.47
Mixed land-use	Mixed land-use entropy	CMAP	0.46	0.01	0.24	0.75
<i>Socio-demographic</i>						
Car free	The proportion of households without a car	ACS	0.25	0.02	0.02	0.69
One car	The proportion of households with one car	ACS	0.44	0.01	0.20	0.72
Two or more cars	The proportion of households with two or more cars	ACS	0.31	0.02	0	0.59
Employment rate	The proportion of adults with work	SLD	0.67	0.02	0.14	1.00
Male/female	The ratio of males to female	ACS	1.02	0.05	0.11	2.43
White	Proportion of white	ACS	0.28	0.07	0	0.89
Crime density	Density of crime	CDP	2.23	2.42	0.23	11.63
Median income	The proportion of workers earning \$1250/month or more	SLD	0.78	0	0.65	0.91

4. Methodology

4.1. Lognormal regression

A lognormal regression assumes that the dependent variable is lognormally distributed, which is mainly used to model continuous variables that are right-skewed. The total trip expense in this study was assumed to follow a lognormal distribution to represent the right-skewed pattern accurately. Lognormal regression assumes that the logarithmic value of the dependent variable behaves linearly with predictors. The probability density function of a lognormally distributed response y is:

$$f(y;\mu,\sigma) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right) \quad y > 0 \quad (2)$$

where μ denotes the mean of the natural log of y , σ is the standard deviation of the natural log of y . The corresponding regression model can be written as:

$$\ln(y_i) = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + \varepsilon_i \quad (3)$$

where β_0 and β_j capture the intercept and the effect of the j -th predictor, respectively, and ε_i is the normally distributed error term.

4.2. Bayesian additive regression trees

BART is a Bayesian statistical approach for regression problems, introduced by (Chipman et al., 2010a), to capture nonlinear relationships between predictors and the response flexibly. In BART, the data are denoted as $\{(y_1, x_1), (y_2, x_2), \dots, (y_n, x_n)\}$ for n observations and p predictors. The model is expressed as:

$$Y = \sum_{i=1}^m T_i(x) + \varepsilon, \varepsilon \sim N(0, \sigma^2) \quad (4)$$

where Y is the predicted value of the dependent variable, and ε is the normally distributed residual term. The prior distribution in the BART model governs both the tree structure and the parameters at each terminal node:

$$P(T_j, M_j) = P(T_j) \prod_{b=1}^{B_j} P(\mu_{jb} | T_j) \quad \#(5)$$

The prior distribution specifies the probability of the tree structure T_j , along with the parameters for the terminal nodes, denoted by M_j . Where B_j is the number of terminal nodes in tree T_j , and μ_{jb} is the parameter assigned to the b -th terminal node of tree T_j . Each tree in BART is regularized by prior distributions that limit tree depth, preventing a single tree from dominating the predictions. The probability of splitting a node at depth d_i is determined by:

$$P_{\text{split}}(i, T_j) = \alpha(1 + d_i)^{-\beta}, \alpha \in (0, 1), \beta \in [0, \infty) \quad \#(6)$$

where d_i is the depth of node i , and α, β are hyperparameters. Reasonable values for α and β , such as $\alpha = 0.95$ and $\beta = 2$, ensure that tree depth does not grow excessively, controlling the model's complexity.

4.3. Lognormal regression enhanced by BART

Contrary to the assumption in the lognormal regression, the normality assumption may not hold in many real-world scenarios. This issue is particularly evident in outcomes that are strictly positive and right-skewed (e.g., cost or duration). Consequently, we propose an Enhanced Lognormal Regression approach that integrates BART with a lognormal likelihood. The combined model captures the nonlinear relationships between predictors and response variables while accommodating the right-skewed nature of the data. The formulation of the LN+BART model is given as follows:

$$P(y|x, T, M) = \prod_{i=1}^n \frac{1}{y_i \sigma \sqrt{2\pi}} \exp\left(-\frac{(\ln y_i - \mu_i)^2}{2\sigma^2}\right) \quad \#(7)$$

Here, μ_i is the predicted value from the BART model, which captures the nonlinear relationship between predictors and the dependent variable. The likelihood function assumes that y_i follows a lognormal distribution, where μ_i is modeled as the output from BART. The Bayesian formulation can be written as:

$$p(\theta | y) \propto p(y | \mu, \sigma^2) p(\mu) p(\sigma^2) \#(8)$$

In this framework, μ is modeled as the output from BART, while σ^2 follows an inverse-gamma prior. The parameters μ and σ^2 are estimated through MCMC, where Gibbs Sampling iteratively updates their values until convergence. The conditional posterior distributions are given as:

$$\mu \sim \text{Normal}(\hat{\mu}, \hat{\sigma}^2) \#(9)$$

$$\sigma^2 \sim \text{Inverse - Gamma}(\alpha, \beta) \#(10)$$

where $\hat{\mu}$ and $\hat{\sigma}^2$ represent the estimated mean and variance of the posterior distribution, while α and β are shape parameters for the inverse-gamma distribution of σ^2 . By integrating BART to predict μ , the Lognormal regression model effectively captures the nonlinear relationships between predictors and a skewed response variable.

4.4. Partial dependence plot

A Partial Dependence Plot (PDP) is a graphical tool commonly used in the BART literature (Martin, 2024). Therefore, this study uses PDP to facilitate the visualization and interpretation of the developed machine learning models. The PDP is used to illustrate the relationship between a specific predictor and the model's predicted output, while averaging the effects of other covariates. We can compute a PDP with:

$$\text{PDP}(X_j) = \mathbb{E}[\hat{f}(X_1, X_2, \dots, X_{j-1}, x_j, X_{j+1}, \dots, X_p)] \#(12)$$

where X_j is the predictor of interest, and the other predictors $X_1, X_2, \dots, X_{j-1}, x_j, X_{j+1}, \dots, X_p$ are fixed or set to their expected values. PDP helps to understand the impact of a specific feature on the predicted outcome while holding other features constant.

5. Results and discussions

5.1. Correlation analysis and multicollinearity assessment

The Pearson correlation analysis revealed important relationships between the response and explanatory variables in Figure 3. Several variables demonstrate significant positive correlations with total trip expenses, notably Divvy trips density, POI density, and commercial area proportion. Conversely, distance to CBD exhibited a strong negative correlation with trip expenses, confirming that e-scooter revenue generation diminishes with increasing distance from the city center. These correlation patterns align with established spatial economic principles, where economic activity typically concentrates in central urban areas and diminishes outward (Caspi et al., 2020; Huo et al., 2021; Jin et al., 2023). The relationships identified in this preliminary analysis informed our subsequent modeling approach and provided initial validation of the variables selected for inclusion.

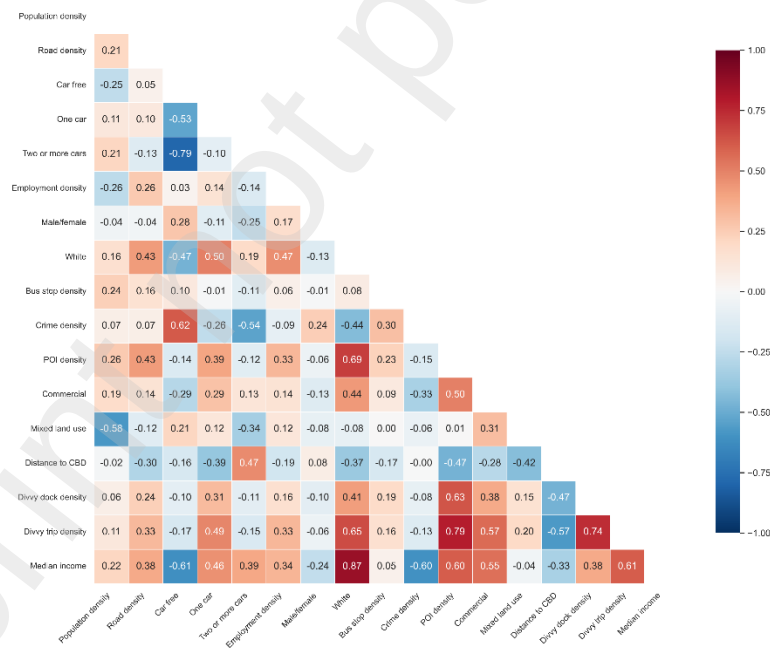


Figure 3. Pearson correlation coefficient of all the explanatory variables

To ensure model stability and reliable parameter estimates, we conducted comprehensive multicollinearity diagnostics using Variance Inflation Factor (VIF) in Table 2.. Through this analysis, variables with correlations exceeding 0.7 or below -0.7 were excluded from subsequent modeling to avoid estimation issues (Ratner, 2009). All retained variables demonstrated VIF values below 5, indicating acceptable levels of collinearity. This diagnostic process ensured that our subsequent modeling efforts would produce stable and reliable parameter estimates, critical for valid interpretation of variable effects and accurate prediction of e-scooter trip expenses.

Table 2. Multi-collinearity analysis based on VIF tests

Feature	VIF
Population density	2.65
Road density	1.58
Car free	4.53
One car	2.46
Male/female	1.16
Crime density	2.84
Bus stop density	1.29
Commercial	2.57
Mixed land use	3.47
Distance to CBD	3.02
Divvy trip density	2.92
Median income	4.51

5.2. Model assessment

Table 3 presents the evaluation results of the models on the test set using three key metrics: MAE, RMSE, and R^2 . The LN+BART model demonstrates superior performance (MAE = 12.36, RMSE = 21.15, R^2 = 0.84) compared to both the BART (MAE = 15.70, RMSE = 28.32, R^2 = 0.71) and RF models (MAE = 16.61, RMSE = 36.83, R^2 = 0.65). This performance hierarchy confirms that the LN+BART method effectively captures the complex relationships between e-scooter trip expenses and the built environment while accounting for the right-skewed nature of economic data.

Table 3. Summary of model performance

Metrics	RF	BART	LN+BART
MAE	16.61	15.70	12.36
RMSE	36.83	28.32	21.15
R^2	0.65	0.71	0.84

Notes: MAE, mean absolute error; RMSE, root mean square error.

5.3. Variable importance

As shown in Table 4, the five most important variables are: median income, POI density, white, distance to CBD, and two or more cars. Socio-demographic variables are the most influential factors in predicting total trip expense, with median income, white, and two or more cars ranking first, third, and fifth, respectively. Land use variables, such as POI density and distance to CBD, ranked second and fourth. Our findings differ from previous studies (Yang et al., 2022), identifying transportation variables as the key predictors. This difference may arise because earlier research focused on traffic volume or link density, whereas our dependent variable, total trip expense, is closely related to social-demographic factors that reflect economic status. Consequently, median income stands out as the most important variable. Some land use variables, like commercial, rank lower in our model, possibly because they often represent trip destinations rather than origins.

Table 4. Variable importance visualization for LN+BART model

Rank	Variables	Importance
1	Median income	0.361
2	POI density	0.307
3	White	0.252
4	Distance to CBD	0.251
5	Two or more cars	0.248
6	Employment density	0.241
7	Divvy dock density	0.191
8	Car free	0.179
9	No car	0.144
10	Population density	0.107
11	Commercial	0.098
12	Road density	0.092
13	Crime density	0.085
14	Bus stop density	0.085
15	Male/female	0.010
16	Mixed land use	0.001

5.4. Nonlinear effects

In this section, we utilize PDP to explain the complicated nonlinear effects of independent variables on the total trip expense. PDP represents a way of visualizing the relationship between an independent and dependent variable while all other variables are held at their mean (Friedman, 2001). Figure 4 illustrates the contribution of independent variables to e-scooter trip expenses based on the LN+BART model. It is

clear from the figure that median income has the most significant impact on the dependent variable. As median income increases from 0.65 to 0.9, the total trip expense rises from 5 to 30. The blue shading around the curve represents the HDI (Highest Density Interval), which reflects the uncertainty around the predicted relationship. A narrow HDI means higher confidence in the model's prediction for a given interval of the independent variable, while a wider HDI implies higher uncertainty. In contrast, variables such as male/female and one car have very small impacts on the dependent variable as their PDPs do not vary much. Figure 6 displays the spatial distribution of the dependent variable and some critical independent variables, thus providing an insightful understanding that deepens the analysis of partial dependence plots (PDPs).

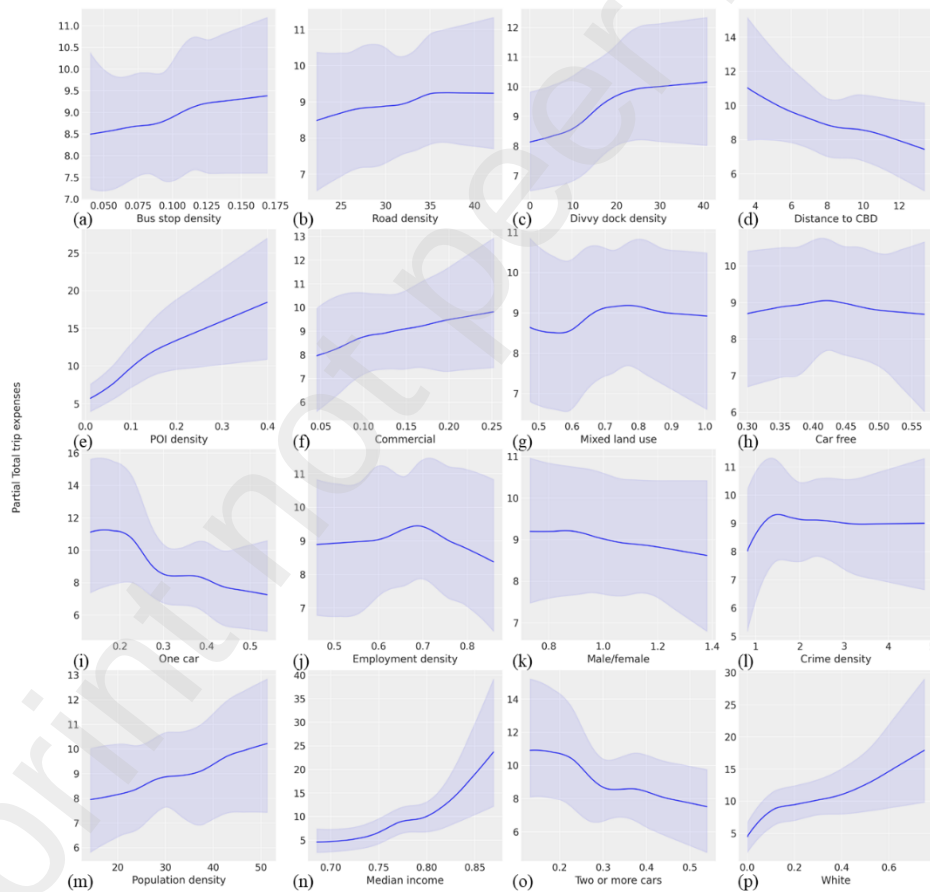


Figure 4. Partial dependence plots of independent variables on e-scooter trip expenses

(1) Transportation infrastructure

As shown in (a)-(c) of Figure 4, the transportation infrastructure variable positively impacts the total trip expense, indicating that well-developed roads and transportation infrastructure enhance the economic benefits of e-scooters. Among these factors, divvy dock density has the most significant effect on the dependent variable. Specifically, as divvy docks density increases from 0 to 20, the total trip expense rises from 8 to 10. This finding suggests that e-scooter operators need not worry about divvy competing for market share. At least according to the 2019 data, the market for micro-mobility remains substantial.

Interestingly, this is contrary to a previous study that used the Chicago e-scooter data (Tuli et al., 2021), where our model reveals a positive relation between the bus stop density and the dependent variable. This could be attributed to the inclusion of rail stations in bus stop counts by Tuli et al. (2021) and their use of total bus stops rather than density. Additionally, their linear methods did not allow for the visualization of PDPs, obscuring complex relationships. Our findings suggest that total trip expense positively correlates with bus stop density, indicating that riders use e-scooters for first/last-mile connectivity to bus stops, which has been similarly observed by the findings of Zuniga-Garcia et al. (2022).

(2) Land use

As shown in (d)-(g) of Figure 4, all three land use variables positively impact the total trip expense except for the Distance to CBD. The decrease in total trip expenses as distance to CBD increases, aligns with our intuitive understanding from Figure 5 (a). Additionally, the positive effects of commercial, POI density, and mixed land use on total trip expense indicate that higher economic activity and diversity in an area correlate with greater profitability for e-scooter operators deploying more vehicles in these regions.

POI density is highly correlated with the total trip expense. As POI density increases from 0 to 0.6, the average predicted total trip expense rises from 5 to 20 before leveling off, as shown in Figure 5 (d). The distribution of POI density closely matches that of total trip expense, except for the northwest area of the study region. Higher POI density in an area indicates a high volume of foot traffic and a greater propensity

for people visiting certain places, like theaters and police stations, to have higher spending power. Consequently, they are more likely to choose e-scooters for transportation. also significantly increases usage

(3) Socio-demographic

As illustrated in Figure 4 (h)-(p), the social-demographic variables median income and white have significant positive effects on total trip expense, consistent with (Tuli et al., 2021). Figure 5 (c) and (e) show that areas with a high proportion of white residents and higher median incomes are predominantly in the eastern and northwestern parts of the city. This observation indicates that the introduction of e-scooters in these regions may yield greater profitability for operators.

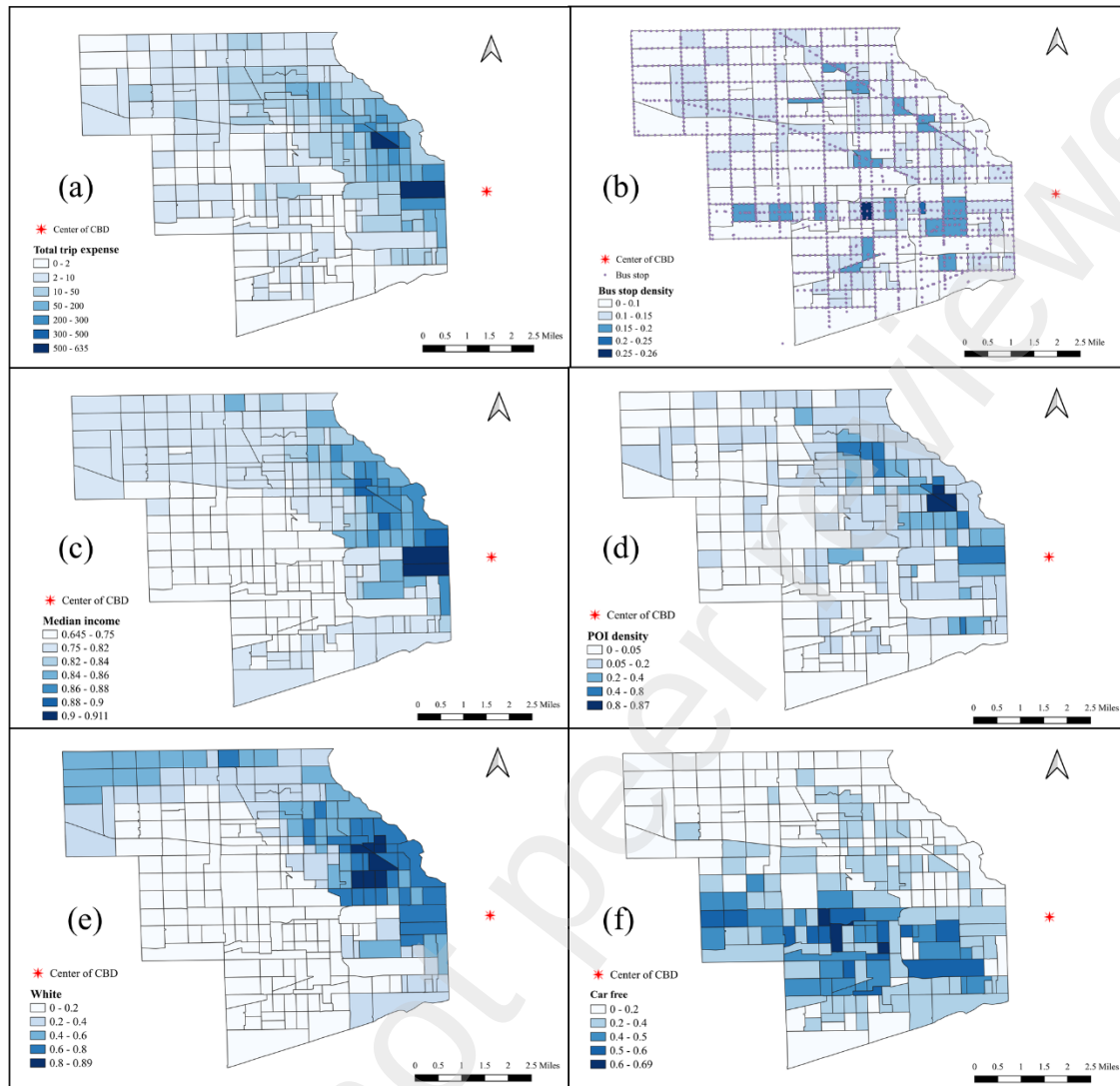


Figure 5. Distribution of e-scooters (a) Total trip expense (b) Bus stop density (c) Median income (d) POI density (e) White (f) Car free

Regarding car ownership, an initial increase in the proportion of car-free households correlates with an increase in total trip expense, but this trend reverses beyond a certain point. This is possibly related to the reliance on other transportation alternatives, such as e-scooters in lower-car areas, while a high share of car-free households may reflect poorer economic conditions, which reduce e-scooter affordability. As shown in Figure 5 (b), (c), and (f), households with the highest proportion of car-free residents are concentrated in the southwest corner of the study area. These areas have low median income, high bus stop

density, and low total trip expense. Similarly, as the proportion of households with two or more cars increases, the preference for personal vehicle use rises, leading to a decrease in total trip expense. Interestingly, unlike previous studies (Tuli et al., 2021), our findings show that total trip expense initially increases with crime density but declines at higher levels. This may be due to the positive correlation between crime density and both population density and bus stop density, initially boosting e-scooter usage. However, higher crime density eventually signals poorer economic and security conditions, leading to a decline in total trip expenses.

6. Strategic implications for stakeholders

6.1. Revenue optimization strategies

Our findings offer actionable guidance for e-scooter operators seeking to optimize revenue generation. The strong influence of median income and POI density, shown in Figure 4(e) and 4(n), suggests that deployment should prioritize areas with higher economic activity and consumer spending power. Figure 5(a), 5(c), and 5(d) illustrate that high-expense areas concentrate in the city's eastern regions, particularly West Town, Near West Town, and areas with elevated median income along the west bank of the North Branch Chicago River.

The nonlinear relationships identified, particularly the plateauing effect of POI density beyond certain thresholds, indicate that operators should consider marginal returns when allocating resources. Rather than concentrating exclusively on areas with the highest POI density, a more efficient strategy may involve ensuring sufficient coverage in areas with moderate-to-high POI density while maintaining adequate service levels in high-income areas, even if they have lower POI concentrations.

6.2. Transportation equity implication

Our results reveal critical transportation equity challenges in e-scooter service provision. Figure 4(h) demonstrates that total trip expenses initially increase with higher proportions of car-free households but decline as this proportion grows further, suggesting that economic barriers—rather than lack of demand—limit e-scooter usage in certain communities. This pattern, paired with the spatial analysis in Figure 5(e) and 6=5(h), indicates that southern and southwestern parts of the study region face particular access challenges despite potentially benefiting most from improved mobility options. These findings highlight a fundamental tension between profit-driven deployment and equitable service distribution that requires deliberate policy intervention.

Addressing these inequities requires a multi-faceted approach centered on economic accessibility and strategic resource allocation. Tiered pricing structures could substantially reduce financial barriers through income-based discounts or subsidies in lower-income, car-free neighborhoods while maintaining baseline service viability. This economic intervention should be complemented by performance metrics that extend beyond revenue to include equity indicators such as spatial trip distribution and demographic diversity of users. Public-private partnerships offer a promising framework wherein operators receive tangible incentives, reduced permit fees, operational subsidies, or regulatory advantages for meeting specific equity targets in historically underserved communities.

Implementation should prioritize specific geographic areas with demonstrated need, particularly those exhibiting high proportions of car-free households combined with lower bus stop density (as shown in Figure 5(b) and 5(f)). These communities could benefit substantially from e-scooters addressing public transit gaps and enhancing mobility options. Previous market-tested initiatives demonstrate the feasibility of such approaches, including Lime's 2020 program that offered 50% discounts in Chicago's southern and western neighborhoods (CBS, 2020). Our model provides a data-driven methodology for identifying census tracts where similar interventions would maximize equity benefits while maintaining essential operational

sustainability, creating a structured framework for balancing commercial viability with social inclusion goals.

To sum it up, the apparent tension between revenue optimization and equity considerations requires a nuanced approach to e-scooter system planning. Rather than viewing these objectives as mutually exclusive, our analysis suggests opportunities for integrated strategies that address both concerns. For instance, the positive relationship between bus stop density and trip expenses indicates that focusing deployment near transit in lower-income areas could enhance equity and generate reasonable revenue through first/last-mile connectivity. Similarly, targeting areas with moderate POI density in underserved communities could provide essential mobility services while maintaining sufficient trip generation to support operational viability. Policymakers can leverage these insights to design regulatory frameworks that encourage optimal fleet distribution across diverse neighborhoods while ensuring adequate service levels in areas that might otherwise be neglected under purely profit-driven deployment models.

7. Conclusions

This study investigated the nonlinear relationships between built environment factors and zone-level e-scooter trip expenses in Chicago through a novel lognormal regression model enhanced by Bayesian Additive Regression Trees (LN+BART). By analyzing over 560,000 trips across 249 census tracts, we integrated 16 variables spanning transportation infrastructure, land use, and socio-demographic characteristics to develop a comprehensive understanding of the economic geography of e-scooter services. This research addresses a critical gap in micro-mobility literature by focusing on the economic dimension of e-scooter operations rather than merely usage patterns, offering insights that balance profitability considerations with transportation equity concerns.

Our analysis revealed several key findings about the determinants of e-scooter trip expenses. First, we identified important nonlinear relationships through partial dependence plots, including threshold effects where trip expenses increase rapidly with median income up to certain points before leveling off. Second, POI density demonstrated a strong positive relationship with trip expenses that plateaued at higher density levels, indicating diminishing returns beyond certain thresholds. Third, the relationship between car ownership and trip expenses was non-monotonic, with initial increases in car-free households associated with higher expenses before reversing at higher levels. Finally, our spatial analysis confirmed that high-expense areas concentrate in the city's eastern regions, particularly in areas with elevated median income and POI density.

From a practical perspective, we propose recommendations tailored to different stakeholders. For e-scooter operators seeking to optimize revenue, deployment strategies should prioritize areas with higher economic activity, particularly regions with higher median income, greater POI density, and proximity to central business districts. The nonlinear effects identified suggest considering marginal returns rather than concentrating exclusively in areas with the highest values of individual predictors. For policymakers addressing equity concerns, we recommend implementing tiered pricing structures with income-based discounts in lower-income neighborhoods, developing performance metrics that extend beyond revenue to include equity indicators, and establishing public-private partnerships where operators receive incentives for meeting equity targets. Urban planners should focus on strategic infrastructure investments in underserved areas with high proportions of car-free households and limited public transit access, as these communities could benefit substantially from enhanced mobility options that address transit gaps.

Methodologically, this study makes significant contributions by introducing the LN+BART approach for analyzing right-skewed economic data in transportation contexts. The integration of a lognormal distribution component with the flexible, nonparametric BART algorithm enables more accurate modeling of economic outcomes. The superior performance of our model compared to traditional approaches (R^2 of

0.84, lower MAE and RMSE) confirms its effectiveness in capturing complex relationships between urban characteristics and economic outcomes. This methodological innovation extends beyond e-scooters to offer transportation researchers a powerful tool for investigating nonlinear relationships across various mobility services where economic outcomes exhibit non-normal distributions.

This research has several limitations that suggest directions for future work. The cross-sectional nature of our analysis captures only a snapshot during Chicago's 2019 pilot program, highlighting the need for longitudinal studies examining how relationships evolve over time, particularly in response to policy interventions or service expansions. Additionally, while our census tract-level analysis provides valuable insights, future research could benefit from more granular spatiotemporal analysis capturing individual trip characteristics and precise origin-destination patterns. Our analysis also focuses on a specific urban context, raising questions about the transferability of findings to cities with different built environments, socioeconomic compositions, or regulatory frameworks. Future research should explore these contextual variations through comparative studies across multiple cities, potentially integrating qualitative methods to enhance understanding of the behavioral mechanisms underlying the statistical relationships identified.

Declaration of generative AI-assisted technologies in the writing process

We used ChatGPT to draft this paper to improve grammar and language. The authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Built Environment Impacts on Zonal E-Scooter Expenses: A Bayesian Machine Learning Approach

Guocong Zhai^a, Ruigan Wang^{a, b}, Hongtai Yang^{a*}, Xiang Liu^a, Miloš N. Mladenović^c, Yandong Tang^a, Huaqiao Mu^d, Xiaobo Liu^a

^a School of Transportation and Logistics, National Engineering Laboratory of Integrated Transportation Big Data Application Technology, National United Engineering Laboratory of Integrated and Intelligent Transportation, Southwest Jiaotong University, Chengdu 611756, China.

^b Tangshan Graduate School, Southwest Jiaotong University, Tangshan 063000, Hebei, China.

^c Department of Built Environment, School of Engineering, Aalto University, Finland.

^d Guolanzhongtian Environment Technology Group, Chengdu, 611756, China.

*Corresponding author

Email: zhaigucong@gmail.com (G. Zhai), wangruigan@my.swjtu.edu.cn (R. Wang), yanghongtai@swjtu.cn (H. Yang), xiangliu@my.swjtu.edu.cn (X. Liu), milos.mladenovic@aalto.fi (M. N. Mladenović), yandongtang92@my.swjtu.edu.cn (Y. Tang), muhuagiao@glzt.com.cn (H. Mu), xiaobo.liu@swjtu.cn (X. Liu)

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CrediT authorship contribution statement

Guocong Zhai: Conceptualization, Methodology, Writing – original draft. **Ruigan Wang:** Methodology, Formal analysis, Writing – original draft. **Hongtai Yang:** Conceptualization, Methodology, Writing – review & editing. **Xiang Liu:** Methodology, Formal analysis. **Miloš N. Mladenović:** Writing – review & editing. **Yandong Tang:** Methodology, Writing – review & editing. **Huaqiao Mu:** Writing – review & editing. **Xiaobo Liu:** Conceptualization, Writing – review & editing.